

Superconducting qubit readout:

An open quantum system perspective

Kacper Kłos

1 Transmon–resonator system

We begin our analysis of the transmon–resonator system with the Hamiltonian given in Eq. (31) of Blais et al. [1]:

$$\hat{H} = 4E_C(\hat{n} + \hat{n}_r)^2 - E_J \cos \varphi + \sum_m \hbar\omega_m \hat{a}_m^\dagger \hat{a}_m, \quad (1.1)$$

where $\hat{n}_r = \sum_m \hat{n}_m$.

We separate this Hamiltonian into three contributions: the transmon Hamiltonian $\hat{H}_t$, the resonator Hamiltonian $\hat{H}_r$, and their interaction $\hat{H}_i$:

$$\hat{H} = \hat{H}_t + \hat{H}_r + \hat{H}_i, \quad (1.2)$$

$$\hat{H}_t = 4E_C \hat{n}^2 - E_J \cos \varphi, \quad (1.3)$$

$$\hat{H}_r = 4E_C \hat{n}_r^2 + \sum_m \hbar\omega_m \hat{a}_m^\dagger \hat{a}_m, \quad (1.4)$$

$$\hat{H}_i = 8E_C \hat{n} \hat{n}_r. \quad (1.5)$$

We examine each contribution in turn.

1.1 Transmon Hamiltonian

To determine how $\cos \hat{\varphi}$ acts on a charge state $|n\rangle$, we first relate the operators $\hat{\varphi}$ and $\hat{n}$. Using $\hat{n} = \hat{Q}/(2e)$, $\hat{\varphi} = (2e/\hbar)\hat{\Phi}$, and $[\hat{\Phi}, \hat{Q}] = i$, we obtain

$$[\hat{\varphi}, \hat{n}] = \frac{1}{\hbar}[\hat{\Phi}, \hat{Q}] = i. \quad (1.6)$$

We can then determine the action of $e^{i\hat{\varphi}}$ on $|n\rangle$. Hadamard’s lemma gives

$$\begin{aligned} e^{i\hat{\varphi}} \hat{n} e^{-i\hat{\varphi}} &= \hat{n} + [i\hat{\varphi}, \hat{n}] + \frac{1}{2}[i\hat{\varphi}, [i\hat{\varphi}, \hat{n}]] \\ &= \hat{n} - 1. \end{aligned} \quad (1.7)$$

It follows that

$$\hat{n}(e^{-i\hat{\varphi}} |n\rangle) = (n-1)e^{-i\hat{\varphi}} |n\rangle. \quad (1.8)$$

Thus, $e^{-i\hat{\varphi}} |n\rangle$ is an eigenstate of $\hat{n}$ with eigenvalue $n-1$. Applying the same reasoning to $e^{i\hat{\varphi}} |n\rangle$ gives $e^{\pm i\hat{\varphi}} |n\rangle = |n \pm 1\rangle$. Together with $\cos \varphi = \frac{1}{2}(e^{i\hat{\varphi}} + e^{-i\hat{\varphi}})$, this determines the action of the transmon Hamiltonian:

$$\hat{H}_t |n\rangle = 4E_C n^2 |n\rangle - \frac{E_J}{2}(|n+1\rangle + |n-1\rangle). \quad (1.9)$$

1.2 Resonator Hamiltonian

The relations needed to expand the resonator Hamiltonian are

$$\begin{aligned}\hat{n}_m &= \frac{C_g}{2e C_m} \hat{Q}_m, \\ \hat{Q}_m &= i \sqrt{\frac{\hbar C_m \omega_m}{2}} (\hat{a}_m^\dagger - \hat{a}_m).\end{aligned}\tag{1.10}$$

Consequently,

$$\hat{n}_m = i \eta_m (\hat{a}_m^\dagger - \hat{a}_m), \quad \eta_m = \frac{C_g}{2e} \sqrt{\frac{\hbar \omega_m}{2 C_m}}.\tag{1.11}$$

We now adopt a single-mode approximation, retaining only one resonator mode. Substituting these expressions into $\hat{H}_r$ and omitting constant terms gives

$$\hat{H}_r = (8E_C \eta^2 + \hbar \omega_0) \hat{a}^\dagger \hat{a} - 4E_C \eta^2 ((\hat{a}^\dagger)^2 + \hat{a}^2).\tag{1.12}$$

To simplify this Hamiltonian, we introduce a Bogoliubov transformation:

$$\hat{\hat{a}} = \cosh(r) \hat{a} - \sinh(r) \hat{a}^\dagger,\tag{1.13}$$

$$\hat{a} = \cosh(r) \hat{\hat{a}} + \sinh(r) \hat{\hat{a}}^\dagger,\tag{1.14}$$

$$[\hat{\hat{a}}, \hat{\hat{a}}^\dagger] = \cosh^2(r) - \sinh^2(r) = 1.\tag{1.15}$$

Substituting into the operator products appearing in $\hat{H}_r$ yields

$$\begin{aligned}\hat{a}^\dagger \hat{a} &= (\cosh(r) \hat{\hat{a}}^\dagger + \sinh(r) \hat{\hat{a}})(\cosh(r) \hat{\hat{a}} + \sinh(r) \hat{\hat{a}}^\dagger) \\ &= \frac{1}{2} \sinh(2r) ((\hat{\hat{a}}^\dagger)^2 + \hat{\hat{a}}^2) + \cosh(2r) \hat{\hat{a}}^\dagger \hat{\hat{a}} + \sinh^2(r), \\ \hat{a}^2 &= \cosh^2(r) \hat{\hat{a}}^2 + \sinh^2(r) (\hat{\hat{a}}^\dagger)^2 + \sinh(2r) \left(\hat{\hat{a}}^\dagger \hat{\hat{a}} + \frac{1}{2} \right), \\ (\hat{\hat{a}}^\dagger)^2 + \hat{\hat{a}}^2 &= \cosh(2r) ((\hat{\hat{a}}^\dagger)^2 + \hat{\hat{a}}^2) + \sinh(2r) (2\hat{\hat{a}}^\dagger \hat{\hat{a}} + 1).\end{aligned}\tag{1.16}$$

For compactness, define

$$A = 8E_C \eta^2 + \hbar \omega_0, \quad B = 4E_C \eta^2.\tag{1.17}$$

Substituting the preceding expressions into $\hat{H}_r$ gives

$$\hat{H}_r = (A \cosh(2r) - 2B \sinh(2r)) \hat{\hat{a}}^\dagger \hat{\hat{a}} + \left(\frac{A}{2} \sinh(2r) - B \cosh(2r) \right) ((\hat{\hat{a}}^\dagger)^2 + \hat{\hat{a}}^2) + \text{const.}\tag{1.18}$$

We choose r so that the terms quadratic in the creation or annihilation operator cancel:

$$\begin{aligned}\tanh(2r) &= \frac{2B}{A} = \frac{8E_C \eta^2}{\hbar \omega_0}, \\ r &= \frac{1}{4} \ln \left(\frac{A + 2B}{A - 2B} \right) = \frac{1}{4} \ln \left(1 + \frac{16E_C \eta^2}{\hbar \omega_0} \right).\end{aligned}\tag{1.19}$$

In the final Hamiltonian, we denote the transformed operator $\hat{\hat{a}}$ by $\hat{a}$ for simplicity. The transformation must nevertheless be accounted for when expressing the interaction term. The resonator Hamiltonian becomes

$$\hat{H}_r = \hbar \omega \hat{a}^\dagger \hat{a} + \text{const.},\tag{1.20}$$

where

$$\begin{aligned}\hbar\omega &= A \cosh(2r) - 2B \sinh(2r) \\ &= \sqrt{A^2 - 4B^2} = \omega_0 \sqrt{1 + \frac{16E_C\eta^2}{\hbar\omega_0}}.\end{aligned}\tag{1.21}$$

Here we have used standard hyperbolic-function identities to obtain ω . In this form, the action of $\hat{H}_r$ on its eigenstates $|j\rangle$ is immediate.

1.3 Interaction Hamiltonian

In addition to the approximation introduced for the resonator, we must account for the change in the creation and annihilation operators under the Bogoliubov transformation:

$$\begin{aligned}\hat{a}^\dagger - \hat{a} &= (\cosh(r) - \sinh(r))(\hat{a}^\dagger - \hat{a}) \\ &= e^{-r}(\hat{a}^\dagger - \hat{a}).\end{aligned}\tag{1.22}$$

As in the preceding subsection, we use $\hat{a}$ to denote the transformed annihilation operator in the final expression:

$$\hat{H}_i = 8iE_C\eta\hat{n}e^{-r}(\hat{a}^\dagger - \hat{a}) = ig(\hat{a}^\dagger - \hat{a}).\tag{1.23}$$

1.4 Total Hamiltonian

Combining the contributions above gives

$$\hat{H} = 4E_C\hat{n}^2 - \frac{E_J}{2}(e^{i\varphi} + e^{-i\varphi}) + \hbar\omega\hat{a}^\dagger\hat{a} + ig(\hat{a}^\dagger - \hat{a}).\tag{1.24}$$

We express this Hamiltonian in the basis of Cooper-pair charge states associated with the Josephson junction and dressed-photon states of the resonator.

References

- [1] Alexandre Blais, Arne L. Grimsmo, S. M. Girvin, and Andreas Wallraff. Circuit quantum electrodynamics. *Reviews of Modern Physics*, 93(2):025005, May 2021. doi: 10.1103/RevModPhys.93.025005.